

| Layer                 | Technology                                                                                            | Why                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Frontend              | <b>React (Vite) + Tailwind CSS</b>                                                                    | Fast setup, component reuse across 3 dashboards, easy styling without writing custom CSS    |
| Routing               | <b>React Router</b>                                                                                   | Role-based protected routes (/admin/*, /staff/*, /student/*)                                |
| State Management      | <b>React Context API</b> (or Zustand if team knows it)                                                | Auth state + role, no need for Redux at this scale                                          |
| Backend               | <b>Node.js + Express</b>                                                                              | Fastest REST API setup, huge community, easy for a team to split routes                     |
| Database              | <b>MongoDB + Mongoose</b>                                                                             | Schema-flexible (good when specs evolve mid-hackathon), fast to prototype vs SQL migrations |
| Authentication        | <b>JWT + bcrypt</b>                                                                                   | Stateless auth, role embedded in token, simple middleware for RBAC                          |
| File Uploads          | <b>Multer</b> (local disk) or <b>Multer + Cloudinary</b> (if you want persistent URLs across deploys) | Handles document uploads (marksheets, ID proof, etc.)                                       |
| Charts/Analytics      | <b>Recharts</b>                                                                                       | Clean React integration, minimal boilerplate for pie/bar/line charts                        |
| Notifications         | <b>In-app</b> (Notifications collection + bell icon) + <b>Nodemailer</b> (bonus, if time allows)      | Skip SMS — needs paid APIs, not worth setup time                                            |
| Deployment (Frontend) | <b>Vercel</b>                                                                                         | One-click deploy from GitHub, free tier                                                     |
| Deployment (Backend)  | <b>Render</b> or <b>Railway</b>                                                                       | Free tier Node hosting, easy env var setup                                                  |
| Database Hosting      | <b>MongoDB Atlas</b> (free tier)                                                                      | No local DB setup needed, works from day one across your team                               |
| Version Control       | <b>Git + GitHub</b>                                                                                   | Branch per module (feature/admin-dashboard, etc.)                                           |

## Supporting Tools

| Purpose               | Tool                                                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------|
| API testing           | <b>Postman</b> or <b>Thunder Client</b> (VS Code extension) — test endpoints before frontend integration |
| Environment variables | <b>dotenv</b> (.env file, never commit it)                                                               |
| API calls (frontend)  | <b>Axios</b> — cleaner than raw fetch for interceptors (auto-attach JWT to every request)                |
| Form handling         | <b>React Hook Form</b> — less boilerplate than manual useState per field, built-in validation            |
| Date handling         | <b>date-fns</b> or plain JS Date — for deadlines/status timestamps                                       |
| Icons                 | <b>Lucide React</b>                                                                                      |